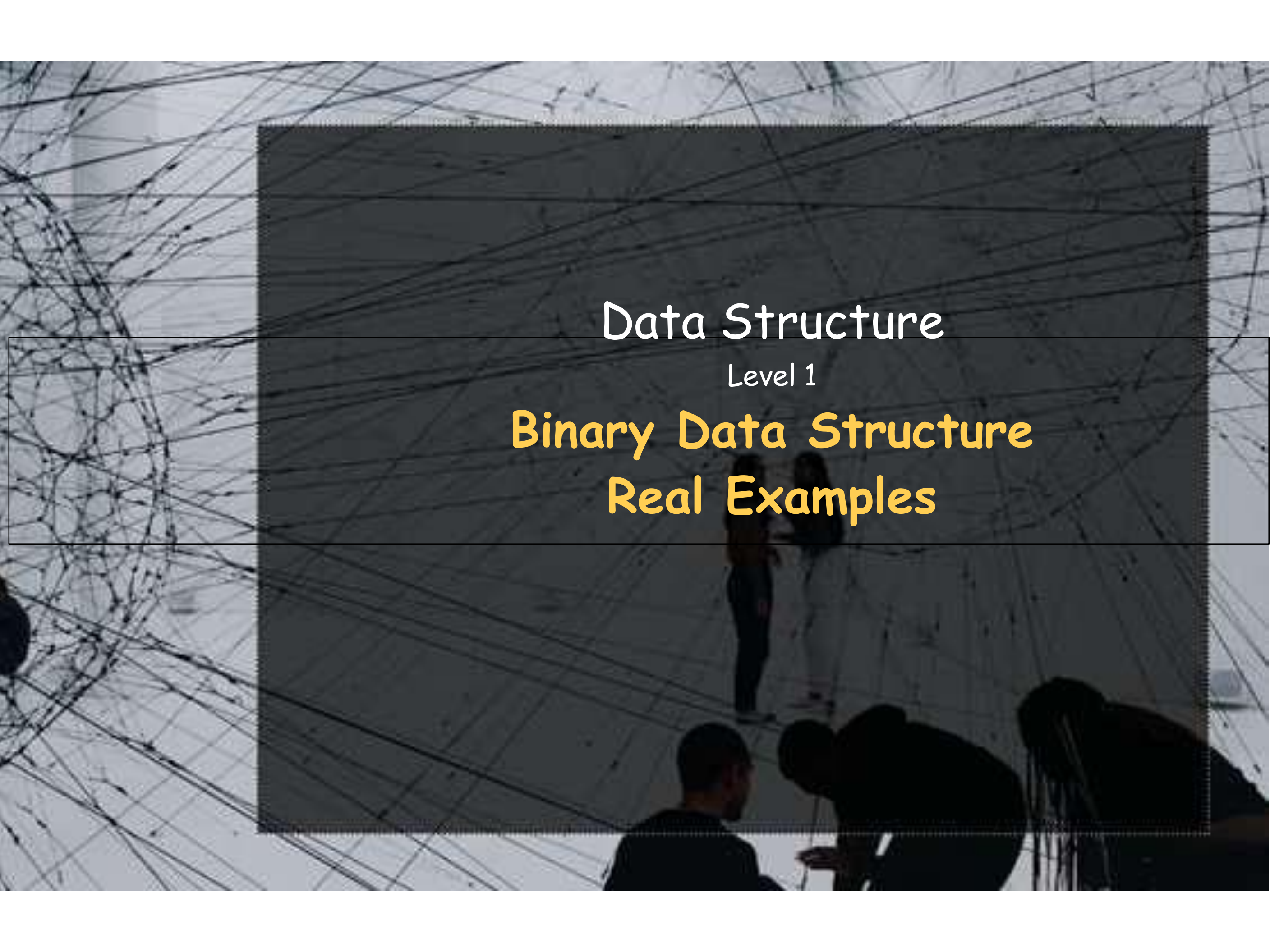


Data Structure

Level 1

Binary Data Structure

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background, resembling a neural network or a data visualization. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark gray rectangular area is overlaid on the right side of the image, serving as a backdrop for the text.

Data Structure

Level 1

Binary Data Structure
Real Examples

You can store data in a number using binary

What does this number mean?

79

Number 79 means:

Permission	Has Permission?	Binary Value
Show Client List		1
Add New Client		2
Delete Client		4
Update Client		8
Find Client		16
Transactions		32
Manage Users		64
Show Login Register		128

= 79

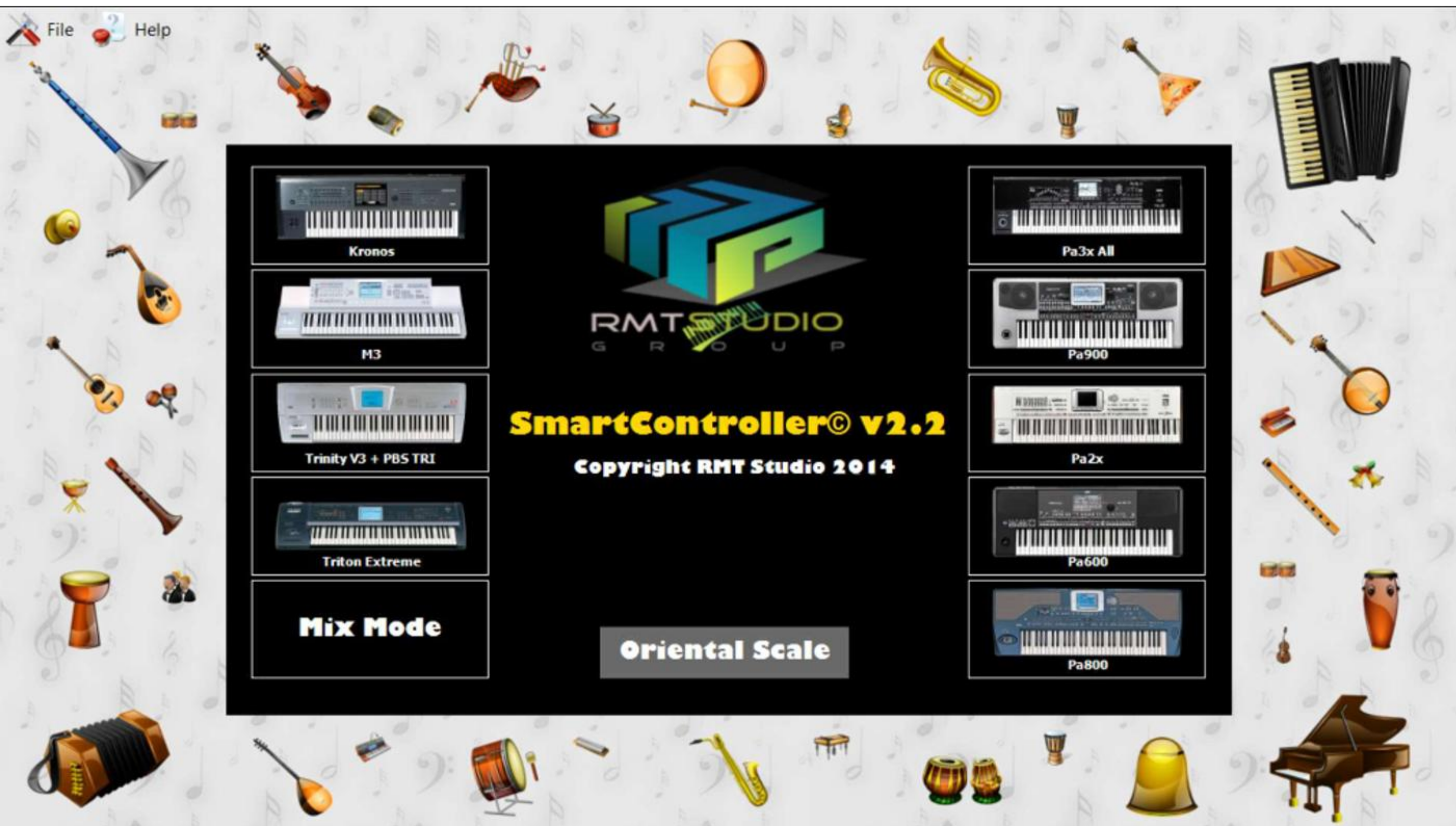
What is Time & Space Function of it? $O(?)$

Permission	Has Permission?	Binary Value
Show Client List		1
Add New Client		2
Delete Client		4
Update Client		8
Find Client		16
Transactions		32
Manage Users		64
Show Login Register		128

If
(PermissionToCheck & UserPermissions =
PermissionToCheck)
yes
Else
No

$O(1)$

Another Practical Example



Another Practical Example

SetList1

SetList2

SetList3

Bass & Arabic String Ens

Detuned PMW Oct +

Bass & Taqsim Kamanja

Tenor Sax T3

Stage Arabic Strings

Square

Trumpet Section

Lead

Mr Oboe RMT

Accordion Master 2

Clarinet Vel. Trill

Bass Kanoun 1 Vel. Tr

Solo1

Bass & Trinity Saz

Bass Nay V3

Bass Oud Vel. Tr

Korg Kronos

Bass & Live Strings Ens.

Gypsy Guitar Stum LR

Nay Small

Phaser Bozuk C2 -C3 Rib

Kamanja Z1

Solo 7 Guitar/Bozuk

Sax Z1 -Y Growl

Bass & E.Saz

Trombone -Y Growl STR

Bass & E.Guitar

Bass Oriental Saz JS -Y

Inter. Organ

Accordion Master 1

Bass & Takseem Guitar

Bass & Alto Sax JS -Y

Old World Choir/Strings

Bass & Arabic St Soft AT

Full Brass Section

Bass & Dudduk JY + Vib.

Solo 3 Distobon

Kelfar Piano + SoftPad 2

Bass & Sweet Keman

Bass & Biaban JY + Vib.

Solo 4 RD

Accordion Musete 9

Bass & Keman Saw

Solo 1 Sync

Solo 5 Phaser

Turkish Ensemble String

Bass & Kelfar Violin

Solo 2 Flanger

Solo 6 Gt. Phaser

Octave

Oct -

Off

Oct +

P1

P2

P3

P4

ري الهوى

جانا الهوى

حب ايه

عسنتياك

انا بعشقتك

حبيبي انت

على بابي واقف فمري

تدعي للنسي

العذاب

بنولس بيك

ايت عمري

في يوم وليلة

قارله الخنجر

اصعب فراق

بعيني على كلمة

لوكي خزين

Pa2X Styles

Oriental Scale

-50

-50

-50

-50

-50

-50

-50

Reset

1

2

3

4

5

Globale Effects

Mfx

Tfx

Ifx

Transpose

b -

0

+ #

P1

P2

P3

P4

Strings

Kawala SADA

Another Practical Example

Korg Kronos

SetList1 **SetList2** **SetList3**

Bass & Arabic String Ens
Stage Arabic Strings
Detuned PMW Oct +
Square
Bass & Taqsim Kamanja
Trumpet Section
Tenor Sax T3
Lead

Octa
Oct -
Off

P1 P2 P3 **P4**

ري الهوى
جانا الهوى
انا بعشقتك
حبيبي انت
العذاب
بنولس بيك
قارله الفتيان
اصعب فراق
بنعيني على كلمة
لوكي حزين

Program Settings

Program Name: Bass & Nay V3
Bank Number: 25
Program #: 11

☒ Image

accordion	accordion_bayan	accordion_button	accordion_chro...	accordion_melo...	alboka	anglais_cor	archlute
arghul	aulochrome	autoharp	baglama	bagpipe	balalaika	bandoneon	bandura
banjo	barytone	bassoon	bau_ken	bawu	bell	bells_jingle	bells_tubular
bifora	biwa	bombarde	boomwhackers	bouzouki	bugle	bugle_keyed	cavaquinho

Cancel Save

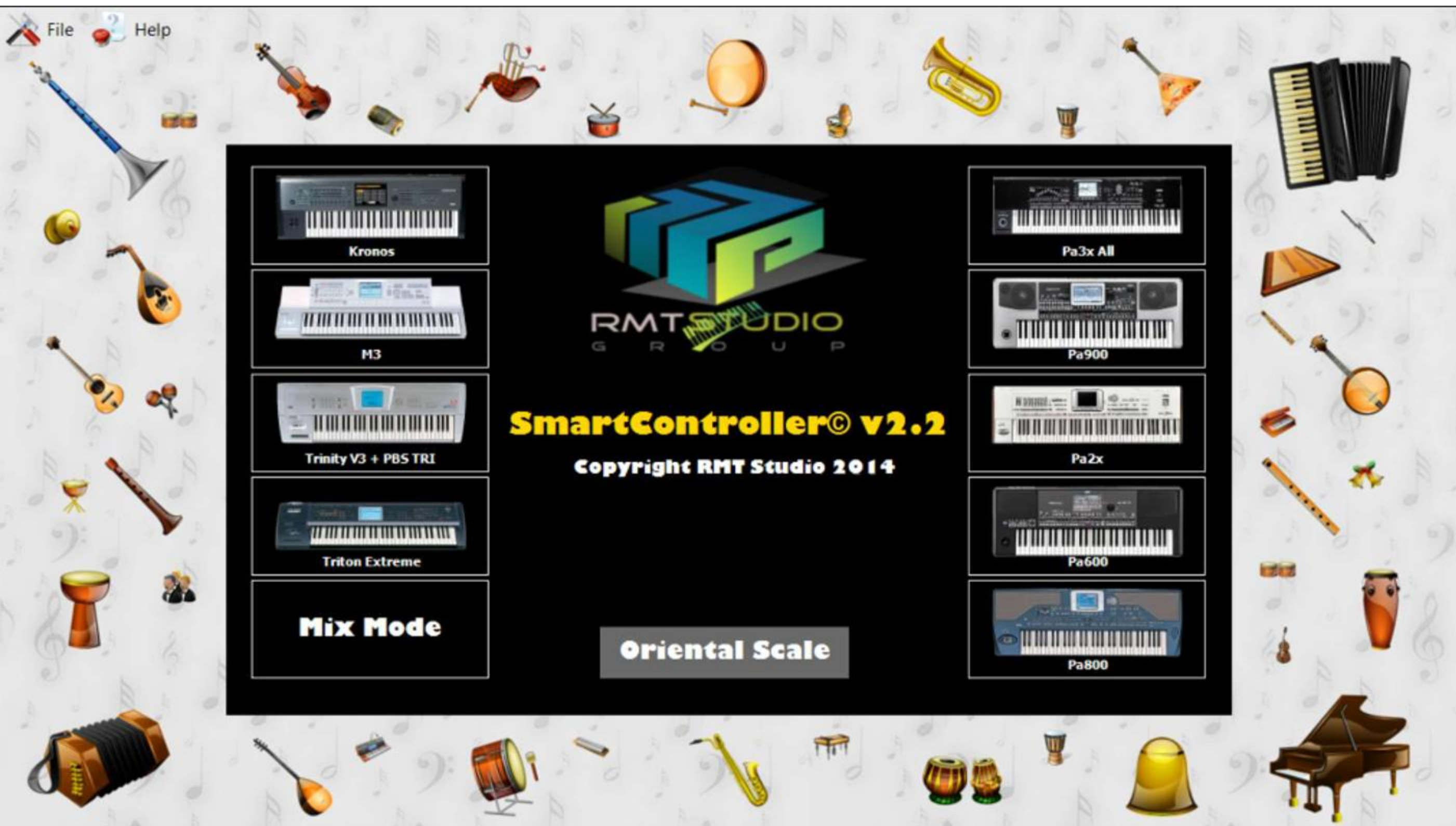
Mfx Tfx Ifx

Accordion Musete 9
Turkish Ensemble String
Bass & Keman Saw
Bass & Kelfar Violin
Solo 1 Sync
Solo 2 Flanger
Solo 5 Phaser
Solo 6 Gt. Phaser

pose
0 + #

Pa2X Sounds

Another Practical Example



DKHSF-PYHQ8-58A8P-GRHCB-BV2ZS

Thank you 😊